

The Continuous Objection

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Abstract

Every approach to the Riemann Hypothesis reaches the same wall from a different direction. The wall is not a collection of separate obstacles. It is one objection, continuous and persistent, that changes form at each approach but does not go away: *you cannot prove where the zeros are without an operator that acts on them, and the operator requires the zeros to act on it*. The argument is circular. This paper names the continuous objection precisely, traces its appearance in four major approaches (spectral, statistical, explicit, functional), and proposes that the self-inhibition conjecture [2] is the correct attempt to break the circle — to show that the system decides its own spectrum from within, without an external operator, without circularity.

1 The Objection

Objection 1 (The continuous objection to RH). Let ρ range over the non-trivial zeros of $\zeta(s)$. To prove RH is to show $\operatorname{Re}(\rho) = \frac{1}{2}$ for all ρ .

Every approach requires an operator $\mathcal{O}$ that acts on the zeros and forces them to the line. But constructing $\mathcal{O}$ requires knowing the distribution of the zeros — which is what we are trying to prove.

The objection is circular and continuous: it does not disappear when rephrased. It reappears in every grammar, wearing the face of that grammar, until the grammar runs out.

2 Four Faces of the Objection

Proposition 2 (Spectral face: Connes). *The spectral triple $(\mathcal{A}, \mathcal{H}, D)$ on the adèle class space $X = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^*$ encodes the zeros as absorption frequencies $[4, 5]$. The trace formula reproduces the Weil explicit formula. The object is correct.*

The objection: to show the zeros lie on $\operatorname{Re}(s) = \frac{1}{2}$, one must show the distribution $W(h) \geq 0$ for all admissible $h \geq 0$. But establishing positivity of W requires controlling the contribution $\sum_{\rho} h(\rho)$ from the zeros — which requires knowing $\operatorname{Re}(\rho) = \frac{1}{2}$. The positivity argument uses RH to prove RH. The circle closes.

Proposition 3 (Statistical face: random matrices). *The zeros of $\zeta(s)$ have GUE statistics: the pair correlation function of the zeros matches the pair correlation of eigenvalues of random*

unitary matrices (Montgomery conjecture, verified numerically by Odlyzko). The resonance [3] between zeros and random matrices is observed. It is precise. It is not a coincidence.

The objection: GUE statistics are a property of the distribution of zeros, not of their location on the line. A zero at $\text{Re}(\rho) = \frac{1}{2} + \epsilon$ for very small ϵ would still produce GUE-like statistics in any finite range. The statistical resonance does not force the line. It is consistent with the line. It does not imply it. Observation is not proof. The circle is statistical: we observe what we would expect to observe if RH is true, which is not the same as observing that RH is true.

Proposition 4 (Analytic face: the explicit formula). *The Weil explicit formula [5] is the resonance equation: zeros (spectral) = primes (arithmetic), in exact balance. The zeros and primes are in correspondence. The correspondence is proved.*

The objection: correspondence is not location. The explicit formula holds regardless of where the zeros are in the critical strip $0 < \text{Re}(s) < 1$. It is a global identity that does not distinguish zeros on the line from zeros off the line. The balance is real. The balance does not force the line. A perfectly balanced scale does not tell you where the weights are — only that they are equal.

Proposition 5 (Symmetry face: the functional equation). *The functional equation $\xi(s) = \xi(1-s)$ (where $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$) forces the zeros to be symmetric about $\text{Re}(s) = \frac{1}{2}$: if ρ is a zero, so is $1-\bar{\rho}$. This means zeros off the line come in pairs $(\rho, 1-\bar{\rho})$ symmetric about the line.*

The objection: symmetry about the line is not the same as lying on the line. A zero at $\frac{1}{2} + \epsilon + it$ has its symmetric partner at $\frac{1}{2} - \epsilon - it$. Both are off the line. Both are symmetrically placed. The functional equation is necessary for RH but not sufficient. It constrains the zeros to the critical strip and forces their symmetry. It does not force them to the line. The symmetry is real. The symmetry does not close the circle.

3 The Structure of the Objection

Theorem 6 (The objection is the Arithmetic Wall). *The continuous objection is not a technical obstacle. It is the Arithmetic Wall [1] appearing in the language of proof attempts.*

Over $\mathbb{F}_q$: the objection does not arise. Frobenius acts on the cohomology of $X/\mathbb{F}_q$ as an external operator with known eigenvalues. The eigenvalues are the zeros. Their location is forced by the purity of Frobenius (Weil II [6]): $|\alpha| = q^{w/2}$ for all eigenvalues α of weight w . The operator exists. The circle does not form.

Over $\mathbb{Q}$: Frobenius does not exist globally. There is no external operator with known eigenvalues. The zeros must locate themselves. But self-location requires an internal mechanism — and constructing that mechanism requires knowing the location. This is the circle. This is the Wall. The objection is the computational face of the missing Frobenius: the Halting problem of $\zeta(s)$.

4 Breaking the Circle

Conjecture 7 (Self-inhibition breaks the circle). *The circle can be broken if and only if $\zeta(s)$ self-inhibits [2]: if the system $(\mathcal{A}, \mathcal{H}, D)$ contains an internal mechanism that forces $\operatorname{Re}(\rho) = \frac{1}{2}$ without requiring knowledge of $\operatorname{Re}(\rho)$ as an input.*

A self-inhibiting mechanism would be: an operator or structural property of $\mathcal{H}$ that implies $\operatorname{Re}(\rho) = \frac{1}{2}$ prior to and independently of knowing the distribution of zeros.

The self-adjointness of D would be such a mechanism: a self-adjoint operator has real spectrum, and real spectrum on this system forces the line. Self-adjointness does not require knowing where the zeros are. It is a property of the operator itself.

To prove RH: prove D is self-adjoint on $\mathcal{H}$. That is the one step that breaks the circle. That is week 10.

Remark 8 (The objection as guide). The continuous objection is not an obstacle to be eliminated. It is a guide.

Each time the objection appears — in the spectral, statistical, analytic, or symmetry approach — it names precisely what is missing: an operator that acts on the zeros without requiring knowledge of the zeros. The objection points, each time, to self-adjointness. The circle breaks exactly when the operator is self-adjoint.

The objection is the shape of the missing mechanism, described negatively: what a proof cannot do is precisely what a proof must do without doing it. Self-adjointness is the positive statement of what the objection demands.

The grammar is exact. The proof is the inequality. The objection is the map to it.

References

- [1] Rincón, D. (2026). The Arithmetic Wall. Preprint.
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- [5] Rincón, D. (2026). Week 09: The Resonance Object. Preprint.
- [6] Rincón, D. (2026). The Weil Conjectures as Proof of Concept for RH. Preprint.